# Angio OMEX Experiment Report — angio_lchem_sweep_20261020_182022

3 combinations | 3, 3, 3 replicate(s) each | sim_time 100.0 MCS

## Experiment Archive

Report OF the experiment archive **angio_lchem_sweep_scan.omex** (7,218 bytes) — the SED-ML/COMBINE description of the whole scan.

| Property | Value |
|---|---|
| Parameter scan | lchem ∈ [100, 500, 1000] |
| Scan points | 3 |
| Replicates / point | 3 |
| Metrics compared | num_clusters, cell_density, largest_cluster_size, vegf_mean |

## Experiment OMEX Contents

| File | Size (B) | Role |
|---|---|---|
| combo_data/lchem_100.csv | 763 | Per-combination replicate-mean metrics (CSV) |
| combo_data/lchem_1000.csv | 718 | Per-combination replicate-mean metrics (CSV) |
| combo_data/lchem_500.csv | 747 | Per-combination replicate-mean metrics (CSV) |
| manifest.xml | 942 | OMEX manifest (COMBINE archive index) |
| metadata.rdf | 1,661 | OMEX Metadata (provenance: creator, license, citation) |
| model/model_summary.json | 2,829 | Bundled model artifact (model/@source resolves here) |
| scan.sedml | 13,549 | SED-ML comparison (scan repeatedTask + plot2D per metric) |
| scan_metadata.json | 4,227 | Scan catalogue (points, combinations, metrics) |

## Reading the SED-ML (comparison)

| plot2D (signal across combinations) | curves (= combinations) |
|---|---|
| num_clusters across combinations | 3 |
| cell_density across combinations | 3 |
| largest_cluster_size across combinations | 3 |
| vegf_mean across combinations | 3 |

## Validation of the Experiment Archive

| Layer | Result |
|---|---|
| Built-in SED-ML | PASS |
| Built-in manifest | PASS |
| biosimulators-utils | PASS (0 errors, 1 warning group) |

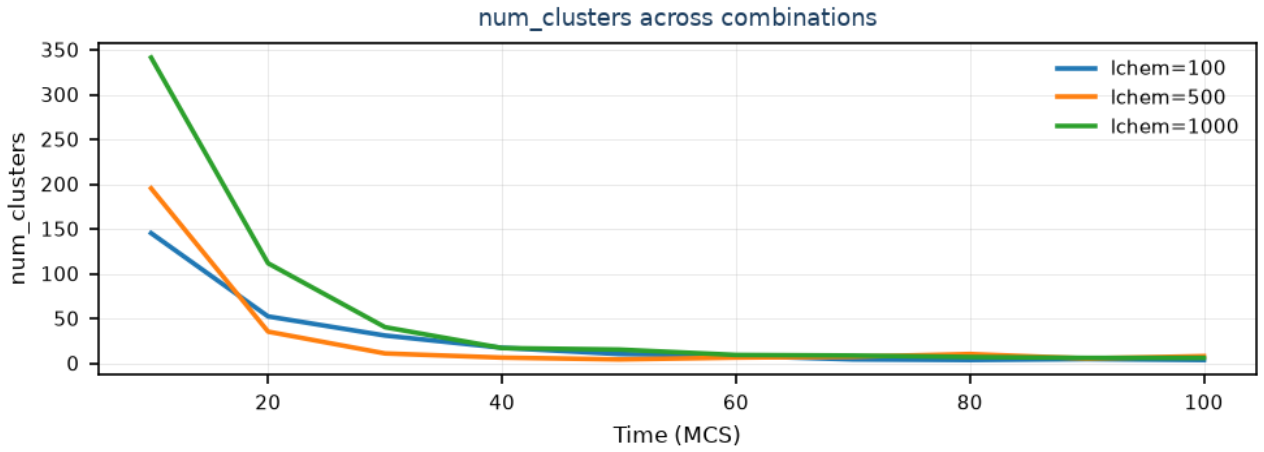
## Provenance (OMEX Metadata)

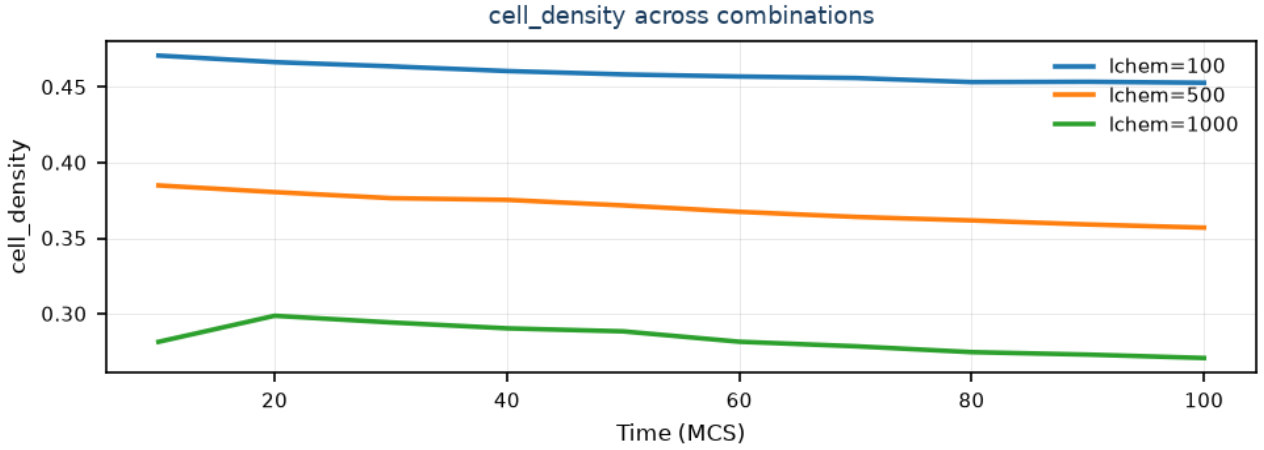
| Field | Value |
|---|---|
| Creator (experimenter) | Joel Vanin |
| Created | 2026-10-21T08:38:12.560429 |
| Modified | 2026-10-21T08:38:12.560429 |
| License | https://spdx.org/licenses/CC-BY-4.0.html |
| Citation | Merks, R.M.H., Brodsky, S.V., Goligorksy, M.S., Newman, S.A., Glazier, J.A. (2006). Cell elongation is key to in silico replication of in vitro vasculogenesis and subsequent remodeling. Developmental Biology 289(1), 44-54. https://doi.org/10.1016/j.ydbio.2005.10.003 |
| Keywords | angiogenesis, parameter scan, cellular Potts model, VEGF |

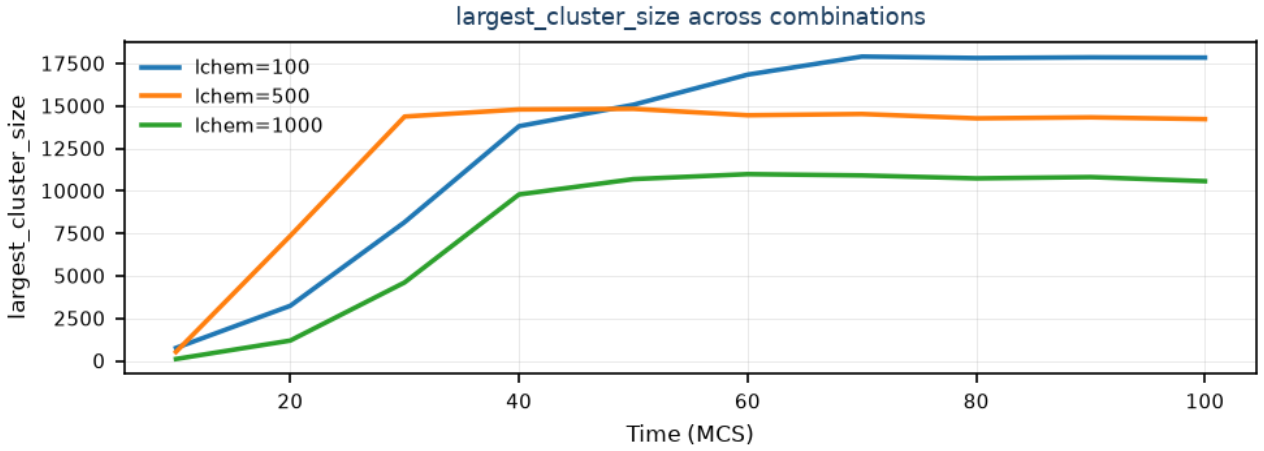
## Summary Metrics (replicate-mean, final step)

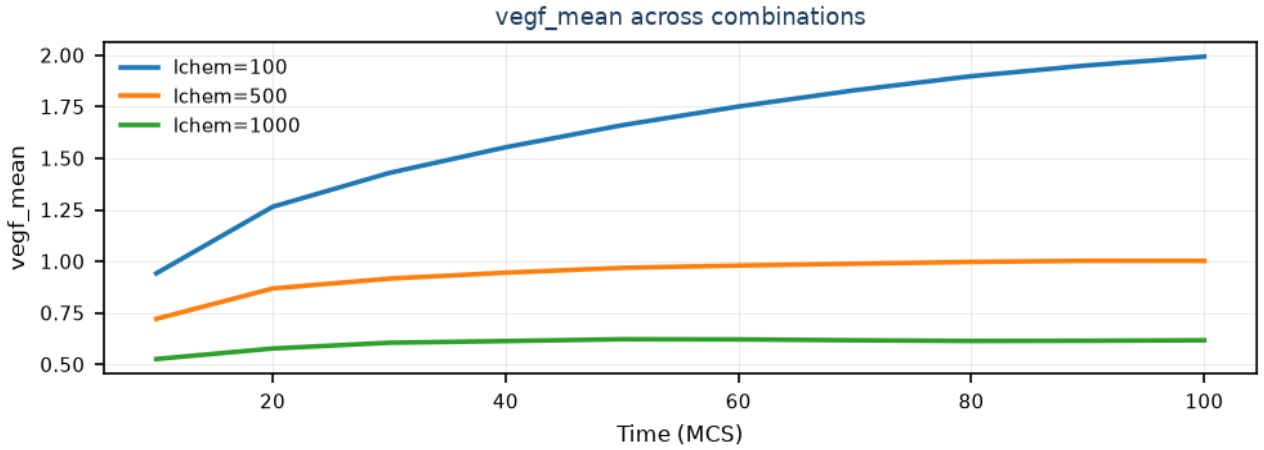
| Combination | num_clusters | cell_density | largest_cluster_size | vegf_mean |
|---|---|---|---|---|
| lchem=100 | 3.67 | 0.453 | 1.78e+04 | 1.99 |
| lchem=500 | 8 | 0.357 | 1.42e+04 | 1 |
| lchem=1000 | 6 | 0.271 | 1.06e+04 | 0.618 |

## Metric Comparisons (one line per combination)

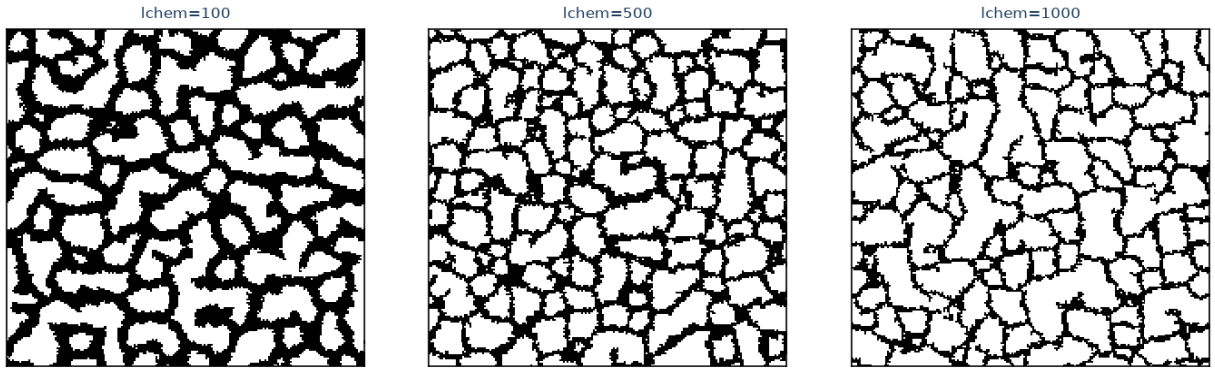





## Final-step EC networks (one per combination)




## Final-step VEGF fields (one per combination)

